Stock O3 profile: QCD $u\bar{u} \to Zggg$

A separate stock-profile measurement. Actual `loop_sm` exports use O3 for all generated MODEL/DHELAS/process/driver objects, normal learned filters and DP-interface timing without the independent stability test. Earlier checked, fixed-point O0 results keep their own input/filter policy; ratios across these pages are not compiler speedups.

Five explicit reduction backends

Table 1: Actual stock `check profile` per-point estimates. Each row is a separate run with its own batch size; FULL–SKIP estimates reduction cost, not a directly measured kernel time. Unavailable/failed results remain explicit.

Reducer	FULL ms	SKIP ms	FULL−SKIP estimate ms	FULL points	FULL wall s	FULL ratio / NinjaRS 8
CutTools	1579.818810	1458.729219	121.089591	50	79.133	×1.089
Ninja	1504.547799	1451.542921	53.004877	50	75.357	×1.037
COLLIER	1771.917629	1406.543589	365.374041	50	88.729	×1.221
NinjaRS	1450.696761	1437.618103	13.078658	50	72.661	×1.000
TIRust	2260.844975	1368.611798	892.233177	50	113.179	×1.558

TIRust FULL is 2260.845 ms versus NinjaRS 1450.697 ms: ratio 1.55845, or 55.8% higher. This remaining runtime gap is explicit; the ratio is descriptive, with no paired confidence interval.

Measured configuration and clock

`CTModeRun=1`, `LoopPoleMode=-1`, `HelicityFilterLevel=2`; normal filter learning is retained. COLLIER's UV and IR controls are both true. These effective cards, selected helicities/filter state, model and all-generated-object O3 compilation records are retained with the raw logs. Installed reducer release builds are unchanged. Timed CTMode1 omits MadLoop direction/rotation checks and whole-amplitude QP rescue. All runs retain 48 contributing helicities out of 96. Timed energy is 1 TeV, verified from the actual FULL driver, external masses and physical card. GNU Fortran 15.2.0; CPU 12 affinity on a shared, unreserved host. Observed O3 compiler commands cover 174 base objects; quiet stock recompiles inherit O3 make_opts/environment/recipe, without directly logged argv. The initial model dependency-race build and successful retry are both retained.

FULL is the log's helicity-sum time per phase-space point: (monotonic child-completion wall minus stock boot time)/actual count. SKIP and boot retain Python wall clocks. Each FULL and SKIP stage contains 50 points. Stock/filter initialization uses a fresh three-point learning startup; compiler time is separate. It is not a library-only microbenchmark or the earlier diagnosed one-point initialization experiment.

Table 2: Initialization is separate from warmed stage estimates. A null initialization denotes unavailable measurement. Accepted DP transport does not establish native-DP arithmetic for TIRust.

Reducer	Initialization s	Status	Observed precision scope
CutTools	47.64849	Completed; unchecked 110	Configured DP; timed routes unknown
Ninja	19.36933	Completed; unchecked 110	Configured DP; timed routes unknown
COLLIER	113.52161	Completed; unchecked 110	Configured DP; timed routes unknown
NinjaRS	16.89482	Completed; unchecked 110	Configured DP; timed routes unknown
TIRust	94.28168	Completed; unchecked 110	Configured DP; timed routes unknown

Accuracy, scope and remaining acceptance

One retained common FULL endpoint has identical six-momentum and normalization words across all five runs. Its ten Born/finite/pole pairwise comparisons pass 3e-6 and 1e-8 (maximum relative residual 1.759e-09). This does not qualify all 50 timed points. Endpoint return code 110 is unchecked and output freshness was not independently instrumented. The separate one-point stability phase is outside timing.

TIRust's DP-input path may use local 106-bit DD or 256/512-bit tensor arithmetic with DP inputs/numerators. Those tiers differ from genuine 113-bit Fortran-QP input transport. Stock timings alone do not assign per-point tensor routes; missing counts must not be read as zero. These are configured-backend timings, not accepted-backend or native-DP-only status proofs.

These are unpaired shared-host stock runs, with no paired confidence interval or main-process acceptance claim. All 66 fresh local numerical lanes and eight main runtime gates remain pending. The upper 95% paired candidate/baseline runtime bound ≤1.05 for every main process is not established. The integration goal remains incomplete.

Source `cda704e28d634e893fabc3ac358d9a27789f93aa` TIRust DSO `d427667a0f2f55d0bd96778c1259ffcf52e3925341890b81c38f5f94e1eb7052`. Exact FULL/SKIP/boot counts, clocks, filters, compiler recipes, failures and initialization records accompany `stock-o3-timings.csv` and the pure reproduction commands. The preceding 47 pages are preserved transitively, without replaying or relabeling their physics. Style: pyAmpliCol 7266c1600f6e, preserved typography and green–white–red timing palette.